Band-1: BSD rank ge 2 – documented L-value pipeline on Cremona 389a1 (LMFDB 389.a1)

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2026-09-11

Band-1: BSD on one curve of rank ≥ 2 — documented L-value pipeline

Honesty stamp. This note does **not** claim the Birch–Swinnerton-Dyer conjecture (BSD) is proved, for this curve or in general. It does **not** claim a Clay Millennium Problem resolution. It does **not** stamp a finished Theory of Everything. It records one well-documented elliptic curve over $\mathbb{Q}$ with **analytic and algebraic rank 2**, lists what a **complete L-value pipeline** would require, and reports **box-computed** Frobenius traces a_p / Dirichlet coefficients a_n with explicit **numerical vs literature / proved** labels.

Curve lock (one object)

Field	Value	Status
Cremona label	`389a1`	standard
LMFDB label	`389.a1`	https://www.lmfdb.org/EllipticCurve/Q/389/a/1
Conductor N	389 (prime)	literature / tables
Minimal model	$y^2 + y = x^3 + x^2 - 2x$	$[a_1, a_2, a_3, a_4, a_6] = [0, 1, 1, -2, 0]$
Discriminant Δ	389	tables
CM	none	tables
Mordell–Weil	$E(\mathbb{Q}) \cong \mathbb{Z} \oplus \mathbb{Z}$	LMFDB / Cremona
Generators (affine)	$(0, 0),\ (1, 0)$	LMFDB; box-checked on model
Algebraic rank r	2	tables (not re-proved tonight)
Analytic rank r_{an}	2	LMFDB / Dokchitser-style tables (**literature**)
Torsion	trivial	tables
Root number / bad prime	split multiplicative at 389; local root number -1	LMFDB local data

Why this curve. Among elliptic curves over $\mathbb{Q}$, conductor 389 is the **smallest** known for Mordell–Weil rank 2 (Cremona; Elkies tables; OEIS A052432). It is the standard rank-2 test object for BSD numerical checks.

What BSD asserts (reminder only)

For $E/\mathbb{Q}$, write $L(E,s)$ for the Hasse–Weil L-function. BSD predicts:

1. $\operatorname{ord}_{s=1} L(E,s) = \operatorname{rank} E(\mathbb{Q})$;
2. the leading coefficient equals the BSD product

$$\frac{L^{(r)}(E,1)}{r!} \stackrel{?}{=} \frac{|\mathrm{Sha}(E/\mathbb{Q})| \cdot \Omega_E \cdot \mathrm{Reg}(E/\mathbb{Q}) \cdot \prod_p c_p}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}.$$

Tonight: we document a pipeline toward (1)–(2) for **this one curve**. We do **not** prove either equality.

Literature BSD ledger for 389a1 (not our computation)

Values below are quoted from LMFDB (accessed 2026-09-11) as a **citation ledger**. They are **not** re-certified by a Dokchitser run on this box (Sage/PARI unavailable).

Quantity	Literature value	Role
r_{an}	2	order of vanishing at $s=1$
$L^{(2)}(E,1)/2!$	$\approx 0.759316500288427$	special value
Real period Ω_E	≈ 4.98042512171011	N'eron period
Regulator Reg	$\approx 0.152460177943144$	MW height pairing
$\prod c_p$	1	Tamagawa (only $p=389$)
$\lvert E(\mathbb{Q})_{\mathrm{tors}}\rvert$	1	torsion
$\mathrm{Sha}_{\mathrm{an}}$ (rounded)	1	analytic Sha

Numerical consistency check (literature arithmetic only):

$$\frac{1 \cdot 4.9804251217 \cdot 0.1524601779 \cdot 1}{1^2} \approx 0.759316500,$$

matching the quoted special value to the printed digits. **This is a ledger identity among published floats, not a proof of BSD and not a proof that** $|\mathrm{Sha}| = 1$**.**

Box computations (this lab)

Environment. `sage` / `gp` (PARI) **not installed** on the shared box. Computations below are **pure Python** point-counts over $\mathbb{F}_p$ and multiplicative extension of a_n.

A. Frobenius traces a_p — proved on box (finite check)

For good $p \neq 389$,

$$a_p = p + 1 - |E(\mathbb{F}_p)|,$$

counting the projective point at infinity. For $p = 389$ (split multiplicative), $a_{389} = +1$ (LMFDB local convention; used only for the Euler factor at N).

Result. First primes (excerpt):

p	2	3	5	7	11	13	17	19	23	29	31	37	41	43	47
a_p	-2	-2	-3	-5	-4	-3	-6	5	-4	-6	4	-8	-3	12	-2

These match the LMFDB modular form `389.2.a.a` coefficients for the listed primes. **Status: box-proved finite arithmetic** (point enumeration), cross-checked against the published q-expansion.

B. Dirichlet coefficients a_n — proved on box (multiplicative recurrence)

Using the weight-2 Euler recurrence

$$a_{p^k} = a_p\, a_{p^{k-1}} - \mathbf{1}_{p\nmid N}\, p\, a_{p^{k-2}},$$

and multiplicativity, we filled a_n for $n \le 5000$. Comparison with the LMFDB q-expansion through $O(q^{20})$:

$$(a_n)_{n=1}^{19} =$$
$$(1, -2, -2, 2, -3, 4, -5, 0, 1, 6, -4, -4, -3, 10, 6, -4, -6, -2, 5).$$

Exact match to LMFDB. **Status: box-proved** relative to the a_p table and the standard newform recurrence.

C. Crude Dirichlet partial sums — numerical only; not L-values

$$S_M(s) = \sum_{n\le M} a_n n^{-s}, \qquad M = 5000.$$

Object	Value	Label
$S_{5000}(1)$	≈ 0.0707	**numerical partial sum only**
$S_{5000}(1.5)$	≈ 0.1348	numerical
$S_{5000}(2)$	≈ 0.3601	numerical
$\sum_{n\le 5000} a_n (\log n)^2 n^{-1}$	≈ 6.67	**not** $L''(E,1)$

Honesty. Without the completed L-function / approximate functional equation (Dokchitser, Rubenstein, etc.), truncated Dirichlet series **do not** certify $L(E,1) = L'(E,1) = 0$ and **do not** equal $L^{(2)}(E,1)/2!$. The partial sum $S_{5000}(1) \approx 0.07$ is **expected noise**, not a contradiction of vanishing: convergence at the central point is conditional and slow. Prefer literature Dokchitser values above.

D. Generator on-curve check — proved on box

Affine points $(0,0)$ and $(1,0)$ satisfy the Weierstrass equation over $\mathbb{Z}$. **Status: elementary verification.** Independence / full Mordell–Weil basis is **not** re-proved tonight (would need canonical heights + saturation); cited from LMFDB/Cremona.

Complete L-value pipeline (what "done" would require)

A **complete documented pipeline** for this curve, suitable for a serious BSD numerical / conditional certificate, needs the following stages. Tonight completed only the items marked **DONE**; the rest are **REQUIRED / NOT DONE on box**.

1. **Model + conductor + minimal discriminant** — **DONE** (locked to `389a1`).
2. **Local data** (Kodaira, Tamagawa c_p, split/nonsplit, root number) — **literature DONE**; independent Tate algorithm **NOT re-run**.
3. **Frobenius a_p table** to height sufficient for approx FE — **partial DONE** ($p \leq 5000$ via a_n fill; extensible).
4. **Modular identification** $L(E,s) = L(f,s)$ for the weight-2 newform of level 389 — **literature** (modularity theorem + Cremona tables); not re-proved.
5. **Completed L-function / approximate functional equation** (Dokchitser `lcalc` / Sage `E.lseries().dokchitser()` / PARI `ellanalyticrank`) $\rightarrow$ numerical $L(E,1)$, $L'(E,1)$, $L''(E,1)$ with rigorous error bounds — **NOT DONE** (no Sage/PARI).
6. **Analytic rank certificate**: prove $\mathrm{ord}_{s=1} L(E,s) = 2$ (vanishing to order 2 + nonvanishing of $L^{(2)}(E,1)$) — **NOT DONE** on box; LMFDB reports $r_{\mathrm{an}} = 2$.
7. **Algebraic rank**: 2-descent / saturation / height matrix $\rightarrow \mathrm{rank}\, E(\mathbb{Q}) = 2$ with generators — **NOT re-proved**; tables say rank 2.
8. **Regulator + real period + Tamagawa + torsion** — literature ledger **quoted**; independent high-precision recompute **NOT DONE**.
9. **Sha**: prove $|\text{Sha}| < \infty$ and compute it (or bound it) — **OPEN in general**; analytic Sha = 1 is **numerical/literature**, not a proof that Sha = 1.
10. **BSD equality**: compare $L^{(r)}(E,1)/r!$ to the BSD product with proved error bars — **NOT claimed**.

Kolyvagin / Gross–Zagier caveat. For analytic rank 0 or 1, much of BSD is known under Heegner / Euler-system hypotheses. For **rank** ≥ 2, including this curve, the equality and finiteness of Sha remain **open** in the strongest sense. That is precisely why Band-1 picks a rank-2 object: the pipeline is **harder**, and honesty matters more.

Artefacts

Path	Contents
`labs/bsd_rank2_389a1/compute_results.json`	a_p, a_n [1, 20], partial sums, environment note
This markdown + PDF	Band-1 pipeline note

Non-claims (hard)

- **BSD is not proved** here.
- **No Clay stamp.**
- **No finished ToE stamp.**
- Analytic rank 2 and $\text{Sha}_{\mathrm{an}} = 1$ are **literature / numerical** unless and until a Dokchitser-class run with certificates is attached.
- Box-proved tonight: **Weierstrass model checks**, a_p **point counts**, a_n **recurrence match to LMFDB** q**-exp through** $n = 19$.

Next executable step (if Sage/PARI lands)

```
E = EllipticCurve([0,1,1,-2,0])   # or EllipticCurve('389a1')
assert E.conductor() == 389
print(E.aplist(100))
print(E.analytic_rank())          # literature expects 2
L = E.lseries().dokchitser()
print(L.derivative(1,2)/2)        # expect ~ 0.759316500...
print(E.sha().an_numerical())     # expect ~ 1
```

Until that (or an equivalent `lcalc` / PARI) runs **on this box** with logged precision, the special-value column stays labeled **literature**, not **box-proved**.

Band-1 item 7 — C.M. Baird / ZoraASI — 2026-09-11. Prefer honesty over fake precision.